

Google Colab

What is it?

Google Colab is an online Google tool for writing and running Python code directly in the browser. It works like an interactive notebook where users can combine text, code, charts, tables, and execution results in a single document.

Its main advantage is that users can start programming without installing Python or setting up a local environment. All they need is a Google account, a notebook, and code cells to run.

How does it work?

A Colab notebook is made of two main types of cells:

- **Text cells:** used for explanations, headings, lists, links, and formulas.
- **Code cells:** used to write Python commands and display the output below the cell.

To run a cell, click the “play” button or use the shortcut **Shift + Enter**.

Simple example

```
grades = [8.5, 7.0, 9.2, 6.8]
average = sum(grades) / len(grades)
print(f"The class average is {average:.2f}")
```

This code calculates the average of a list of grades and immediately shows the result in the notebook.

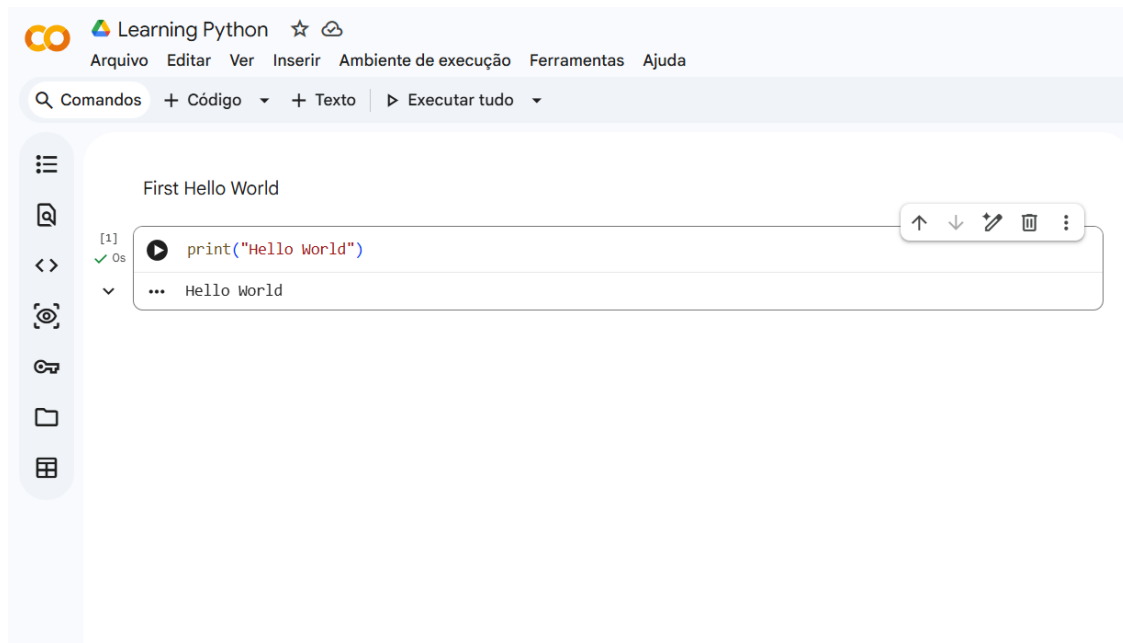
Main advantages

Feature	Benefit
Browser-based use	No initial installation is required.
Google Drive	Makes it easy to save and access files.
Sharing	Allows collaboration with other people.
Ready-to-use libraries	Includes common data science packages.
GPU and TPU	Helps with machine learning tasks.

When should you use it?

Colab is useful for programming classes, data analysis, chart creation, quick Python tests, and machine learning studies. It is also helpful for creating reproducible reports, since explanation and code stay together in the same file.

The image below shows an example related to Google Colab and helps illustrate the visual interface discussed in this document.



Conclusion

Google Colab is a simple and accessible option for learning, testing, and sharing Python code. It stands out because it combines programming, documentation, and results in a collaborative cloud-based environment.